FDRM VIX Project: Study Guide (Compact)

Oct 8, 2026 · @Subham Agrawal

1. The project on one page

The question. Can a long-term Nifty investor buy crash protection cheaply by timing it with India VIX? Using daily data from January 2020 to April 2026, the answer is yes, but only as rare-event insurance, not as a way to beat the market.

The assignment asked for	What we did	Report section / slides
Implied vs realised volatility and the VRP	Compared India VIX with the volatility Nifty delivered over the next 21 trading days	Section 3 / slide 4
India VIX: construction and sentiment use	Explained NSE's formula, re-implemented it, measured how VIX reacts to Nifty	Section 4 / slides 5–6
VIX regimes vs Nifty returns	Low (VIX < 13), Normal (13–20), High (> 20); tested returns and volatility in each	Section 5 / slide 7
Premiums around 4 June 2024; a low-VIX hedge sold into spikes; a backtest by regime	Event study, a rule-based put hedge, and a daily backtest against six alternatives	Sections 6–9 / slides 8–16

The pipeline. Load daily Nifty and India VIX → compute returns, realised volatility and regimes → measure the VRP and regime statistics → price options with Black-Scholes using VIX → simulate seven strategies day by day → stress-test the result → save every number to `output/results.json`. The report and deck are written from that file.

Five facts to remember.

1. India VIX was 2.2 vol points above the volatility that followed, on 78% of days. Options are usually overpriced.
2. The overpricing was largest (4.0 points) right after shocks, which is the time to sell protection, not buy it.
3. Nifty was three times as volatile in high-VIX regimes (30%) as in low ones (9%), but average returns did not differ significantly.

4. Before the 2024 election, option prices doubled over six weeks. Nifty fell 5.9% on result day, almost exactly the move VIX had priced, and VIX collapsed 29% the next day.
5. The proposed hedge (3-month put bought only when VIX < 13, sold into spikes) cut the worst fall from 38% to 22% and returned 13.9% a year against 11.1% unhedged. Almost all of that came from the 2020 COVID crash; from July 2020 it cost about 0.5% a year.

Files (shared `fdrm-vix` folder): `fdrm_vix_analysis.py` and the run notebook `.ipynb`; `data/` (two raw CSVs); `output/` (every table plus `results.json`); `figures/` (8 charts); `fdrm_vix_presentation.pptx` and `.pdf` (17 slides, built by `build_deck.js`). The full-length study guide, with every code block quoted, is here.

2. Volatility, the risk premium and India VIX

Volatility is the annualised standard deviation of returns, in percent. A 16% volatility means a typical move of about 16% a year, 16 ÷ √12 ≈ 4.6% a month, or 16 ÷ √252 ≈ 1% a day (about 252 trading days a year; variance grows with time, so volatility grows with its square root).

- **Realised volatility (RV)** is measured from past prices: the standard deviation of daily log returns $r = \ln(P_t / P_{t-1})$ over 21 trading days, × √252 × 100.
- **Implied volatility (IV)** is read from option prices: the volatility that makes Black-Scholes match the market price. It is the market's forecast plus whatever extra people pay for protection.
- **The volatility risk premium (VRP)** is the gap: VRP = India VIX today − RV over the next 21 trading days. VIX is compared with the future because it is a forecast of the next 30 calendar days.

Why the VRP is usually positive. Option sellers are selling crash insurance and charge more than the expected payout, like any insurer. Investors who own stocks happily pay it, because crashes hurt most when they can least afford it. On the rare days a crash arrives, realised volatility shoots above VIX.

What we found (2020–26). VIX averaged 17.3, forward RV 15.1: a VRP of +2.2 points, positive on 78% of days, Newey-West t = 2.8. By regime it was 2.1 (Low), 1.4 (Normal) and 4.0 (High): after a shock, fear stays high while actual volatility is already calming. VIX correlates 0.86 with the past 21 days' volatility but only 0.61 with the next 21, so it reacts more than it predicts. **Implication:** a hedger who buys options all the time loses money; the edge is in when you buy.

India VIX is NSE's measure of expected Nifty volatility over the next 30 calendar days, launched in 2008 using the CBOE method behind the US VIX and computed in real time from Nifty option quotes.

India VIX	Expected 1-day move (÷ 16)	Expected 1-month move (÷ 3.5)	Regime
11	±0.7%	±3.2%	Low
16	±1.0%	±4.6%	Normal
25	±1.6%	±7.2%	High
60	±3.8%	±17.3%	High (March 2020)

These are one-standard-deviation moves: the market expects a move that size or smaller about two-thirds of the time.

How NSE builds it. (1) Take near-month and next-month Nifty options. (2) Find the forward F from put-call parity; K_0 is the first strike at or below F. (3) Use only out-of-the-money options (puts below K_0, calls above) at bid/ask mid-prices Q(K). (4) For each expiry compute the model-free variance below. (5) Interpolate the two to exactly 30 days. (6) VIX = 100 × √(30-day variance).

$$\sigma^2 = (2/T) \cdot \Sigma\ (\Delta K_i\ /\ K_i^2) \cdot e^{RT} \cdot Q(K_i)\ -\ (1/T) \cdot (F/K_0 - 1)^2$$

"Model-free" means no pricing model is assumed: the fair value of future variance equals a portfolio of OTM options weighted by $1/K^2$. That weight gives low-strike puts the most influence, so when investors pay up for crash protection, VIX rises even if at-the-money volatility does not. **Our check:** the script pushed a synthetic option chain through the formula. With every option at 15% it returned exactly 15.0 (the code is right); with 15% ATM plus put skew it returned 16.1. VIX is part volatility forecast, part price of crash insurance.

VIX as a sentiment gauge. (1) It moves against the market: +2.3% on an average down day, −1.9% on an up day, correlation −0.56 (the *leverage effect*). (2) It clusters: calm and fearful periods last weeks to months, which is why regimes are useful. (3) It mean-reverts: every spike decayed back to 12–16. (4) It prices known events in advance, then collapses once the news is out. **What it does not tell you:** direction. High VIX means a big move is expected, up or down. India VIX futures exist but trade thinly, so VIX is the signal and Nifty options are the instrument.

3. Option pricing and statistics you need

Puts and protection. A put with strike K pays max(K − S, 0) at expiry. Owning Nifty plus a put puts a floor under losses. A put 5% below spot (a "95% put", 5% out of the money) is cheaper because the first 5% of any fall is not covered, like an insurance deductible. Nifty options are European and cash-settled.

Black-Scholes-Merton prices every option in the project:

$$P = K \cdot e^{-rT} \cdot N(-d_2) - S \cdot e^{-qT} \cdot N(-d_1)$$
$$d_1 = \left[\ln(S/K) + (r - q + \sigma^2/2) \cdot T \right] / (\sigma \cdot \sqrt{T}), \quad d_2 = d_1 - \sigma \cdot \sqrt{T}$$

S = Nifty close; K = 95% of spot rounded to 50; T = calendar days ÷ 365; r = 6.5% (about the 91-day T-bill); q = 1.3% (Nifty dividend yield); N = normal probability. Calls come from **put-call parity**, C − P = S·e^(−qT) − K·e^(−rT).

Skew and spread. Real OTM puts trade at higher IV than ATM options because everyone wants crash protection. The model: put IV = VIX + 0.6 points per 1% OTM (so +3 for a 95% put), buy at +0.5 and sell at −0.5 points for the bid/ask. At VIX 12 the hedge put is bought at 15.5% IV.

What the hedge put costs (Nifty 24,000, 22,800 put):

India VIX	1-month put	3-month put
12 (Low)	52 pts (0.22% of spot)	204 pts (0.85%)
16 (Normal)	110 pts (0.46%)	345 pts (1.44%)
25 (High)	281 pts (1.17%)	702 pts (2.92%)

The same protection costs 3.5 to 5 times as much at VIX 25. That is reason one to buy only when VIX is low.

Greeks (3-month 22,800 put at VIX 12): **delta** −0.19 (gains 0.19 pt per 1-pt Nifty fall, growing towards −1 in a crash); **gamma** positive (gains accelerate: convexity); **vega** +33 pts per vol point (VIX 12 → 25 adds about 430 pts with Nifty unchanged, so a crash pays twice, through delta and vega); **theta** −2.1 pts a day against −2.8 for the 1-month put that costs a quarter as much. Slower decay is reason two for the 3-month tenor.

The ATM straddle (ATM call + ATM put) prices the expected size of a move: straddle ≈ 0.8 · σ · √T · S. At VIX 14 a 30-day straddle costs about 3.2% of spot; at VIX 24.6 about 5.6%.

Term structure caveat. VIX is a 30-day measure but the hedge is a 3-month option. Normally 3-month IV is above 30-day IV (so real puts cost more in calm markets); in crises it is below (so they sell for less). Using VIX for both probably flatters the strategy. This is

the main limitation.

Statistics, and why each was used.

- **Log returns** add up over time and are close to % returns for small moves; the code converts back with e^r − 1 when reporting a move.
- **Mean, median, sd, hit rate (share positive), 5th percentile** (a simple bad-case measure) summarise each regime.
- **Newey-West t-statistics.** Forward 21-day returns on consecutive days share 20 of 21 days, so 408 low-VIX days are nowhere near 408 independent observations. Ordinary standard errors would overstate certainty. Newey-West (HAC) errors widen them; the code regresses the series on a constant, so the coefficient is the mean and the HAC t is its Newey-West t-statistic, with lags = horizon (1, 21, 63).
- **Kruskal-Wallis** (rank-based, robust to fat tails) tests whether groups share a distribution: same-day returns by regime, p = 0.66, no difference.
- **Levene** tests equal variance: p < 0.001, volatility clearly differs (9%, 13%, 30%). *VIX regimes sort risk, not return.*
- **VIX beta:** regressing the daily log-VIX change on the Nifty return gives −2.7, so a 1% fall comes with a 2.7% VIX rise.
- **Mincer-Zarnowitz regression** RV = a + b · VIX tests forecast bias (unbiased: a = 0, b = 1). Result a = 1.05, b = 0.81, R^2 = 0.38: VIX is informative but biased upward, which is the VRP.
- **Predictive regression:** next-21-day return on VIX gives +0.18% per VIX point (t = 4.3), mostly the 2020 rebound. Not a timing rule.

Two limits. Six years contain three or four real stress episodes, so crash-driven statistics rest on few events. The cut-offs (13, 20) and strategy thresholds were chosen on the same data (in-sample choice, or *data snooping*); the sensitivity grids show conclusions don't hinge on the exact numbers, but only new data can confirm them.

4. The data: source, checks, cleaning

Source. Both series are Yahoo Finance data. Yahoo, NSE and Kaggle all refused connections (HTTP 403) from the cloud machine the analysis ran on, so the files were downloaded with `git clone` from public GitHub repositories that store the same Yahoo series:

File	Content	Series	Downloaded from
`nifty50_daily_raw.csv`	4,554 days, 17 Sep 2007 – 13 Apr 2026	`^NSEI`	github.com/kalilurrahman/NIFTY_50_STOCK_DATA
`india_vix_daily_raw.csv`	1,627 days, 2 Jan 2020 – 24 Aug 2026	`^INDIAVIX`	github.com/tanya459/india-vix-market-data-analysis

If asked "why GitHub?": the numbers are Yahoo's and were checked against NSE; GitHub was only the delivery route.

Checks. On 4 June 2024 the files show Nifty at 21,884.50 and VIX at 26.75, both matching NSE. VIX's highest close is 83.61 on 24 March 2020, the known COVID record. No duplicate dates; Adj Close equals Close (indices are not adjusted). 16 Nifty days (several 1 Januarys, the Saturday budget session of 1 Feb 2025) have no VIX row and drop out of the join.

Cleaning, in code order. (1) Read Nifty, keep Open/High/Low/Close, drop duplicate dates, sort. (2) Read VIX with `encoding="utf-8-sig"` to strip an invisible byte-order mark that would otherwise break the Date column. (3) Rename the Power BI-style columns ("Sum of Close" → VIX). (4) Compute Nifty returns and volatility on the full Nifty history *before* joining, so a missing VIX day never distorts a return. (5) Inner-join on date. Result: **1,536 trading days, 2 Jan 2020 to 13 Apr 2026.**

Why option prices are modelled. NSE's daily F&O bhavcopy files hold every option's real prices, but they were blocked too. So every premium is a Black-Scholes price at VIX plus skew. Results show what options would have cost at VIX-consistent prices, not traded prices.

Getting more data on your own laptop (these sites are not blocked there): `yfinance` (`yf.download(["^NSEI", "^INDIAVIX"], start="2008-01-01")`); NSE's historical index and India VIX pages; **NSE F&O bhavcopy** for real option prices (packages such as `jugaad-data` can fetch them); the **Nifty 50 Total Returns Index** on niftyindices.com to include dividends; RBI's DBIE for T-bill yields.

Key references. NSE, *India VIX* white paper; Cboe, *VIX White Paper*; Black & Scholes (1973) and Merton (1973); Newey & West (1987), *Econometrica*; Carr & Wu (2009) and Bollerslev, Tauchen & Zhou (2009), *Review of Financial Studies*, on the variance risk premium.

5. The 4 June 2024 election study

The event. Lok Sabha votes were counted on Tuesday 4 June 2024. Exit polls on Saturday 1 June predicted a large majority; markets reacted on Monday 3 June. The result was a surprise: the ruling party lost its single-party majority. A textbook high-uncertainty event: known date, unknown outcome.

Method. Benchmark "normal" = January–April 2024 (median VIX 13.8, 30-day ATM straddle 3.2% of spot, median daily move 0.56%). Track two modelled prices daily from April to July: the **30-day ATM straddle as % of spot** (always 30 days left, so it isolates volatility) and a **real contract**, the NIFTY 27-June-2024 22,500 put (includes time decay and Nifty's level).

Date	What happened	Nifty	VIX	Straddle (% spot)	22,500 put (pts)
April (low)	Calm		10.2	about 2.4	
31 May	Last close before exit polls		24.6	5.6	545
3 Jun	Exit polls: big majority	+3.25%	20.9	4.8	217
4 Jun	Result: no single-party majority	−5.93% (−8.5% intraday)	26.8	6.1	903
5 Jun	Relief rally, vol crush	+3.36%	18.9	4.3	335
7 Jun	Near normal		16.9	3.9	112

The pattern. (1) **Build-up:** VIX more than doubled over six weeks while Nifty stayed between 22,000 and 23,000. (2) **False relief:** exit polls cut VIX 15% and the put lost 60% the day before it was needed. (3) **Shock:** the put was worth 4.2× its previous close. (4) **Crush:** VIX fell 29% the next day; within three days prices were near normal.

The move the market expected. VIX ÷ √252 spreads the month's variance evenly (24.6 → 1.55% a day), but the risk sat on one day. Since variances add, assume 29 normal days at the Q1-2024 median VIX of 14.4 plus one event day:

$$\sigma_{event} = \sqrt{\left[(VIX_{pre}^2 \times 30 - VIX_{base}^2 \times 29) / 365 \right]}$$

With the 31 May VIX of 24.6 this gives 5.8%; with 3 June's 20.9, 4.4%. The actual move was 5.9%. The market priced the *size* well, not the direction.

Other shocks ("move vs implied" = |move| ÷ previous VIX/√252):

Day	Nifty	VIX before → after	Straddle before → after	Move vs implied
COVID, 23 Mar 2020	−12.98%	67.1 → 72.0	15.3 → 16.4	3.3×
Ukraine, 24 Feb 2022	−4.78%	24.5 → 32.0	5.6 → 7.3	3.2×
Election result, 4 Jun 2024	−5.93%	20.9 → 26.8	4.8 → 6.1	4.6×
Yen carry, 5 Aug 2024	−2.68%	14.3 → 20.4	3.3 → 4.7	3.0×
US tariffs, 7 Apr 2025	−3.24%	13.8 → 22.8	3.2 → 5.2	3.8×
Normal day (median)	0.47% abs.	14.7	3.4	0.5×

The election was the only shock that arrived with VIX already high (because it was scheduled); the 2024–25 unscheduled shocks hit with VIX near 14, when protection was cheap. Across the 20 biggest one-day VIX jumps, VIX rose 28%, Nifty fell 3.2% and the straddle rose 1.3% of spot.

Lesson for the hedge. Buy before the build-up, sell into the build-up or the shock, never after the crush. The proposed strategy bought a 21,350 put on 1 April 2024 for 195 points, sold half on 27 May (VIX 23) for 177 and the rest on 5 June for 95. It lost a little, because Nifty closed above 21,350 even on result day: insurance only pays when the fall is deeper than the deductible.

6. The hedge design and the backtest

The rules.

Rule	Setting
Instrument	Nifty put, strike 5% below spot (rounded to 50), on 100% of the portfolio, 3 months to expiry
Entry	First trading day of each month, only if India VIX closes below 13
Roll	If VIX is still below 13 next month, sell the old put and buy a fresh 3-month one
No new buying	VIX 13 or above on the roll date: keep the existing put, buy nothing

Rule	Setting
Monetise half	Sell 50% the first day VIX closes at 22 or above
Monetise the rest	Sell the rest once VIX closes 20% below its highest close since crossing 22
Proceeds	Reinvest in Nifty the same day ("buy the dip" with the payout)

Why each choice, and what was rejected.

- **Put, not a put spread.** Selling an 85% put cuts the cost but stops protecting below −15%. COVID fell straight through it: the always-on spread had a −30% drawdown against −20% for the plain put. VIX futures are too illiquid.
- **5% OTM.** ATM did better here (15.3%, −18.6%), 10% OTM worse (13.0%, −24.9%). 5% OTM is the conventional middle ground.
- **Buy below 13** (about the 30th VIX percentile). Not knife-edge: thresholds 12–15 gave 12.2%–15.7% a year; buying regardless of VIX gave 5.9%–8.5%.
- **3-month tenor. This replaced the first design.** The first version used 2-month puts. Its January 2020 put expired on 2 March, a week before the crash accelerated, and VIX stayed above 13 at every later roll, so it owned nothing during the crash (−36% drawdown). A 3-month put bought on 2 January 2020 lives to 1 April. 1- and 2-month versions returned about 10.4% with −36% to −39% drawdowns; 3- and 6-month versions about 13.7% with about −22%.
- **Hybrid monetisation.** Five rules were compared (3-month puts):

Rule	CAGR	Max DD	COVID window	CAGR from Jul 2020
Hold to expiry	17.2%	−17.2%	−6.3%	15.3%
Trailing (sell all on a 20% fade)	16.5%	−17.2%	−6.3%	15.3%
Hybrid (proposed)	**13.9%**	**−22.0%**	**−21.1%**	**14.9%**
Ladder (half at 22, rest at 30)	11.9%	−33.9%	−33.9%	14.8%
Sell all at 22	11.0%	−35.9%	−35.9%	14.4%

Why the hybrid when two rules beat it. Holding won here because the one crash kept going after VIX first hit 22: on 28 February 2020 (VIX 23, Nifty 11,202) Nifty still had another 32% to fall. In a short spike the opposite happens: the 2024 put worth 177 in late May expired worthless on 1 July if held, while the hybrid got back about 70% of its cost. On the day VIX crosses 22 nobody knows which kind of spike it is, so the hybrid locks in

half and lets half ride. Without COVID, all five rules land within about 1% a year of each other. Staggered tranches (a third of the hedge bought each month) did worse (11.8%, −32.7%) because only a third was bought before VIX left the low regime in early 2020.

How the backtest works. It replays 1,536 days (6.28 years). Wealth starts at 1.0, all in Nifty, held as index *units*; there is no cash account. Buying a put sells Nifty units to pay; selling one buys units with the proceeds. Each put covers the whole portfolio value. Roll dates are the first trading day of each month and expiries are synthetic (a roll date 1, 2, 3 or 6 months later). Options are priced with Black-Scholes at VIX + skew, bought at +0.5 and sold at −0.5 vol points, marked daily at mid, and worth max(K − S, 0) at expiry. Each day: (1) settle expiries, (2) check monetisation triggers, (3) on a roll date buy or roll, (4) record value = units × Nifty + options. All trades are at the daily close using that day's VIX close, slightly optimistic about execution.

The seven strategies. Unhedged; always a 1-month 95% put; always a 1-month 95/85 put spread; always a 3-month 95% put rolled monthly (the proposal's instrument without timing); VIX-timed 3-month put held to expiry; the proposal (timed + hybrid); timed + sell all at 22.

Metrics. CAGR = (end ÷ start)^(1/years) − 1. Volatility = sd(daily returns) × √252. Sharpe = (CAGR − 6.5%) ÷ volatility (CAGR instead of mean return: slightly conservative, same for all). Sortino uses only the downside deviation. Max drawdown = worst fall from a running peak. Calmar = CAGR ÷ size of max drawdown. Hedge cost: every option cash flow is written to a ledger, converted to % of the portfolio that day and summed per year into *premium paid*, *recovered* and *net cost* (negative = options made money); *time hedged* = share of days holding a put.

By regime, two ways. *At purchase:* each option's whole life (premium out, money back) goes to the regime on the day it was bought. *Day by day:* each day's strategy return minus the unhedged return is summed by the previous close's regime, then annualised.

7. Results, honestly read

For each result, know what it supports and what it does not. Examiners reward the second part as much as the first.

VRP. +2.2 points on average, 4.0 after shocks (section 2). *Supports:* buying options all the time loses money, most of all right after a shock. *Does not support:* "always sell options". The 22% of days with a negative VRP include the crashes, when realised volatility beat VIX by up to 64.5 points (5 March 2020).

VIX regimes and Nifty returns.

Regime	Days	Same-day vol	Worst day	Next 21 days: mean (NW t)	Next 63 days: mean	Next 63 days: 5th pct
Low (below 13)	408	9.0%	−1.4%	+1.1% (1.8)	+3.9%	−9.0%
Normal (13 to 20)	737	13.2%	−2.2%	−0.2% (−0.2)	+0.9%	−14.6%
High (above 20)	391	30.1%	−13.9%	+3.4% (3.8)	+9.5%	−7.4%

Supports: regimes sort risk (Levene $p < 0.001$). The worst forward outcomes started in Normal, as VIX rose from low levels, so protection has to be in place before the regime changes. *Does not support:* market timing. Same-day returns do not differ (Kruskal-Wallis $p = 0.66$), and the high-regime rebound is mostly the 2020 recovery.

The backtest (2 Jan 2020 to 13 Apr 2026, wealth starts at 1.0):

Strategy	CAGR	Vol	Sharpe	Max DD	Net hedge cost p.a.	Time hedged	CAGR from Jul 2020
Unhedged	11.1%	18.3%	0.25	−38.4%	0	0%	15.4%
Always 1M 95% put	8.3%	13.2%	0.14	−19.6%	3.0%	99%	9.6%
Always 1M 95/85 spread	7.3%	15.0%	0.05	−30.2%	3.6%	99%	10.1%
Always 3M 95% put	6.3%	11.7%	−0.02	−17.9%	4.9%	97%	7.6%
VIX-timed 3M, hold	17.2%	14.6%	0.74	−17.2%	−4.4%	47%	15.2%
VIX-timed 3M, hybrid (proposed)	**13.9%**	**15.4%**	**0.48**	**−22.0%**	**−2.2%**	**45%**	**14.9%**
VIX-timed 3M, sell all at 22	11.0%	17.7%	0.25	−35.9%	0.2%	43%	14.4%

Negative net cost means the options made money. The proposal paid 2.9% of the portfolio a year in premium and got 5.1% back.

Where the money came from. The proposal made 22 trades; only 5 made money and 3 were monetised early. Two did most of the work:

- **January 2020.** Bought at VIX 11.5 for 93 points (0.8% of the index). Half sold on 28 February at VIX 23 for 546 points; the rest on 31 March at VIX 64 for 3,050 points, 33 times the cost.
- **January 2026.** Bought on 2 January at VIX 9.5 (strike 25,000) for 134 points. Nifty fell 15% to 22,331 by 30 March. Half sold on 9 March at VIX 23 for 1,113 points; the other half expired on 1 April worth 2,321. The hedge cut that fall's drawdown from 15% to 10%.

By regime. *At purchase:* puts the proposal bought (always in Low VIX) returned a net +2.2% of the portfolio a year over their lives; always-on monthly puts lost 0.7%, 1.7% and 0.6% a year on those bought in Low, Normal and High VIX. *Day by day:* the proposal cost 2.1% a year against unhedged on low-VIX days (the premium) and earned 2.0% a year on normal days and 6.3% on high-VIX days.

The bottom line.

- **Robust:** buying only in low VIX beats buying always, at every threshold, skew, spread, tenor and rule tested, and from July 2020 too (14.9% against 7.6% to 10.1%).
- **Robust:** always-on protection is expensive: 3% to 5% a year net, 3 to 5 points of CAGR.
- **Fragile:** the size of the gain. Nearly all of the edge over unhedged is one cheap put alive in March 2020. From July 2020 the proposal returned 14.9% against 15.4% unhedged, about 0.5% a year of insurance cost, and its post-2020 maximum drawdown (−17.2%, the slow 2024–25 correction) matched unhedged.
- **Fragile:** tenor and monetisation rule. 2-month puts, or selling everything at VIX 22, would have missed most of COVID.
- **So:** present it as cheap crash insurance, about 0.5% a year in calm years, that pays hugely in a crash that starts from calm. Not as a way to beat the market.

8. Presenting: likely questions, and ideas to build on

A good answer admits the limitation first, then says why the conclusion still stands.

"Why not real option prices?" NSE option history was not reachable, so premiums are Black-Scholes prices at India VIX plus skew and a spread. VIX itself comes from real option prices, so the level is market-based; skew and term structure are assumptions. Skew from 0 to 1 point per 1% OTM and spreads up to 1 point did not change the ranking.

"VIX is 30-day. Can it price 3-month puts?" Only approximately, and it probably flatters the strategy: calm-market 3-month IV is above VIX (puts cost more), and in a spike it rises less (they sell for less). It is the main limitation.

"Aren't 13 and 20 data-mined?" They are about the 30th and 80th VIX percentiles of this sample, so yes, in-sample. That is why thresholds 12 to 15 were tested: all gave 12.2% to 15.7% a year, 4 to 10 points above buying regardless of VIX.

"Insurance should cost money. How did yours make money?" One put bought in January 2020 for 0.8% of the index sold for 33 times its cost in the COVID crash. From July 2020 the hedge cost about 0.5% a year, which is what insurance should cost.

"So it's just COVID?" The size of the gain, yes. But timed buying beat always-on hedging in every test including after COVID, and the January 2026 put cut that fall's drawdown from 15% to 10%.

"Holding did better. Why the hybrid?" Holding won because the one crash kept deepening after VIX hit 22. In a short spike like the 2024 election, a held put gives everything back; the hybrid recovered 70% of its cost. On the day VIX crosses 22 nobody knows which kind it is.

"Does high VIX mean the market will fall?" No. VIX prices the size of moves, not the direction. After high-VIX days Nifty rose 3.4% on average over the next month.

"Why Newey-West?" Consecutive 21-day returns share 20 days, so they are not independent; ordinary errors overstate significance. Lags equal the horizon.

"Your VIX replication gave 16.1, not 15." That is the point: flat 15% vols return exactly 15.0, so the formula is right; adding put skew gives 16.1 because of the $1/K^2$ weights. VIX includes the price of crash protection.

"Could you have profited from the election?" Only by owning protection before the build-up. Puts bought on 31 May were overpriced, lost 60% after the exit polls, and only paid on result day; VIX fell 29% the next morning.

"Why did the put spread do worse?" It stops paying 15% below spot; COVID fell about 38%, straight through it.

"How would a fund run this?" First trading day of each month: if VIX < 13, buy listed Nifty puts 5% OTM, about 3 months out, on the portfolio's notional. Sell half at VIX 22, the rest on a 20% fade. It runs off a daily data feed.

Ideas to build on, by impact.

1. **Real option prices** from NSE F&O bhavcopy files: invert Black-Scholes for real IV (`scipy.optimize.brentq`), measure real skew and term structure by regime, and re-run the backtest at settlement prices. This removes the main limitation.
2. **Out-of-sample test, 2008–2019**, with the rules frozen (13, 22, 20% fade, 3 months, 5% OTM). It covers 2008, 2011, 2013, 2015–16 and 2018.
3. **Dividends:** use the Nifty 50 Total Returns Index.
4. **Realistic costs:** next-day execution, NSE's real expiry calendar, STT, brokerage, fees, GST, wider spreads for 3-month options.
5. **Smarter signals:** a percentile-based threshold (VIX below its own 1-year 30th percentile), a VRP-based entry, GARCH or HAR-RV forecasts against VIX, or a hidden Markov model for regimes.
6. **Other structures:** collars, far-OTM tail puts, or hedging 50% instead of 100%.
7. **Block bootstrap** of 2020–26 to get a distribution of outcomes and answer "was it just COVID?".
8. **Event study across many events** (budgets, RBI and Fed decisions) to see whether pre-event protection is over- or under-priced.

9. **A live Streamlit dashboard** showing today's regime and the rule's signal: a good demo.

9. The code, condensed

Run it. In the `fdrm-vix` folder: `pip install pandas numpy scipy statsmodels matplotlib`, then `python fdrm_vix_analysis.py` (about 30 seconds). It writes 8 charts to `figures/`, every table to `output/`, and all quoted numbers to `output/results.json`. The `.ipynb` is the same script split at its `# %%` markers (jupytext format) and already executed. To change an assumption, edit the constants at the top and re-run.

Map of the script (725 lines, runs top to bottom):

Lines	Block	Main outputs
1–72	Imports, folders, assumptions, chart style	
75–102	Load and align the data	`nifty_vix_merged.csv`
105–158	Black-Scholes, India VIX formula check	`vix_construction` in results
161–196	Volatility risk premium	fig2, `vrp_by_regime.csv`
199–273	Regimes and Nifty returns	fig1, fig3, `returns_by_regime.csv`
276–383	Election event study	fig4, fig5, `election_window.csv`, `event_comparison.csv`
386–508	Backtest engine `run()`	
511–619	Metrics, 7 strategies, hedge cost, regimes, episodes	`backtest_performance.csv`, ledgers, fig6, fig7
651–725	Robustness grids, heatmap, save results	fig8, `sensitivity_*.csv`, `results.json`

The assumptions (lines 41–51). Every modelling choice is a named constant, so a reviewer can see and change it:

```
R = 0.065            # risk-free (approx. 91-day T-bill)
Q = 0.013            # Nifty dividend yield
SKEW_PER_PCT = 0.6   # extra put IV per 1% OTM (5% OTM -> +3 pts)
HALF_SPREAD = 0.5    # buy at IV+0.5, sell at IV-0.5
LOW_VIX, HIGH_VIX = 13.0, 20.0   # regime cut-offs
SPIKE_1, SPIKE_2 = 22.0, 30.0    # monetisation triggers
TRAIL = 0.20         # sell the rest 20% below the post-spike VIX peak
TENOR = 3; PUT_MONEYNESS = 0.95
```

Loading and the forward-volatility trick (lines 78–96):

```
vix_raw = pd.read_csv(DATA / "india_vix_daily_raw.csv", encoding="utf-8-sig",
parse_dates=["Date"])
nifty["ret"] = np.log(nifty.Close).diff()
nifty["rv21_trail"] = nifty.ret.rolling(21).std() * math.sqrt(252) * 100
nifty["rv21_fwd"] = nifty.ret[::-1].rolling(21).std()[::-1].shift(-1) *
math.sqrt(252) * 100
nifty["fwd21"] = nifty.Close.shift(-21) / nifty.Close - 1
df = nifty.join(vix, how="inner").dropna(subset=["Close", "VIX"])
df["regime"] = pd.cut(df.VIX, [0, LOW_VIX, HIGH_VIX, 1e9], labels=["Low",
"Normal", "High"])
```

- `utf-8-sig` strips the invisible byte-order mark that would otherwise rename the first column.
- `rolling(21)` only looks backwards. To get volatility over the *next* 21 days, reverse the series (`[::-1]`), roll, reverse back, then `shift(-1)` so the window starts tomorrow. Value on day t = sd of returns t+1 … t+21.
- `shift(-21)` pulls the price 21 rows ahead onto today's row: the forward return. The last 21 rows are NaN and are dropped where used.
- Returns are computed on the full Nifty history *before* the join, so a missing VIX day never stretches a return over two days.
- `pd.cut` bins VIX into (0,13], (13,20], (20,∞) and labels them.

Black-Scholes and the VIX check (lines 116–158). `bs_put` is the formula from section 3 written with NumPy, so it prices a whole array of strikes or days at once; `np.maximum(T, 1e-8)` avoids division by zero at expiry. `bs_call` uses put-call parity. `model_free_var` builds a strike grid from 60% to 140% of spot in 50-point steps, finds F and K_0, picks OTM puts below K_0 and calls above (average of both at K_0), and applies the NSE sum with `dK = np.gradient(Ks)`. `india_vix_formula` computes it for 16- and 44-day expiries and interpolates to 30 days. Fed a flat 15% chain it returns 15.0; with put skew, 16.1.

Newey-West in one line (used for the VRP and every regime mean):

```
nw = sm.OLS(v.vrp.values, np.ones(len(v))).fit(cov_type="HAC", cov_kwds=
{"maxlags": 21})
```

Regressing a series on a column of ones gives its mean as the coefficient; `cov_type="HAC"` makes statsmodels compute Newey-West standard errors, so `nw.tvalues[0]` is the overlap-corrected t-statistic. The regime loop (lines 202–212) does the same per regime and horizon with `maxlags` = 1, 21 or 63, then `stats.kruskal` and `stats.levene` test the groups.

Election study (lines 281–329). `straddle_pct` prices a 30-day ATM put plus call on a spot of 1.0, so the answer is directly % of spot. The 27-June 22,500 put is priced each day with days-to-expiry ÷ 365 and `put_iv` (VIX + skew). The event-day move is `sqrt((pre**2*30 - base**2*29)/365)` with `base` = Q1-2024 median VIX. The cross-event table uses `df.index.get_loc(d)` to take the day before and the day of each shock.

The backtest engine `run()` (lines 417–508). State: `units` of Nifty held (wealth starts at 1.0, so `units = 1 / first close`), a list `pos` of open options, and a `trades` ledger. Four inner helpers:

```
def iv(vixl, S, K, side):          # VIX + skew + side*half_spread, side +1
buy, -1 sell, 0 mid
def price(p, d, S, vixl, side=0):  # Black-Scholes, or max(K - S, 0) once T
<= 0
def close_pos(p, d, S, vixl, frac, reason, regime):
    cash = p["sign"] * q * px;  units += cash / S      # sale proceeds buy
Nifty units
def open_pos(d, S, vixl, K, months, sign, regime, frac=1.0):
    V = units * S + value of open options;  qty = frac * V / S   # hedge 100%
of the portfolio
    cash = -sign * qty * px;  units += cash / S        # premium is paid by
selling units
```

There is no cash account: every option cash flow converts to or from Nifty units at that day's close. That is the "reinvest proceeds" rule and the "fund premiums by selling Nifty" rule in two lines. Each day the loop does:

1. **Settle expiries:** any option whose `exp` ⩽ today is closed at intrinsic value.
2. **Strategy logic.** `always_put` and `always_spread` open new 1-month positions on roll dates. For the `timed_*` strategies, monetisation runs first. In `timed_hybrid`, every day with VIX ⩾ 22 updates `p["peak"]`; the first such day sells half (`monetise_1`, flag `sold1`); once VIX ⩽ 0.8 × peak the rest is sold (`monetise_trail`). Then, on a roll date with VIX < `low`, the old put is sold (`roll_sell`) and a fresh `tenor`-month put is bought at strike `round(S * 0.95 / 50) * 50`.

3. **Record:** value = units × S + mark-to-market of open options at mid.

Roll dates are the first trading day of each month; `next_roll(d, k)` returns the roll date k months later, which serves as a synthetic expiry. **The `low=1e9` trick:** "Always 3M put, rolled monthly" is just `run("timed_hold", low=1e9)`; VIX is always below 10^9, so the timed rule buys every month. Same instrument, no timing: a clean test of the timing alone.

One trade, traced (`ledger_s5_timed_hybrid.csv`, trade 0). 2 Jan 2020: Nifty 12,282, VIX 11.49 < 13, so buy the 11,650 put expiring 1 April at IV 11.49 + 3.1 + 0.5 = 15.1% for 92.9 points; the premium, 0.76% of the portfolio, is paid by selling Nifty units. 28 Feb: VIX 23.24 ⩾ 22, sell half at 545.8 (IV at bid). 31 Mar: VIX 64.41 is more than 20% below the 83.61 peak of 24 March, so sell the rest at 3,050.5. On 23 March the hedged portfolio was down 20.6% against 38.0% unhedged.

Metrics and hedge cost (lines 511–553). `metrics()` turns a value series into CAGR, volatility, Sharpe, Sortino, max drawdown (`val / val.cummax() - 1`), Calmar, worst day and worst 21 days. `STRATS` maps each of the seven names to a `run()` call. Each ledger's cash is divided by that day's portfolio value to get % of the portfolio, then summed and divided by 6.28 years into premium paid, recovered and net cost per year.

By regime and episodes (lines 562–619). Cost by regime groups each ledger by `open_regime` (regime when the option was bought) and by `regime` (regime on the cash-flow date). Day-by-day attribution subtracts the unhedged daily return and groups by *yesterday's* regime (`bt.regime.shift()`), so a crash day is counted in the regime that preceded it. The episode table compares value at the start and end of six named windows.

Robustness (lines 654–696). Three loops simply call `run()` with different arguments: buy threshold {12–16, always} × first trigger {20, 22, 25, 30}; skew {0, 0.6, 1.0} × half-spread {0, 0.5, 1.0}, plus tenor {1, 2, 3, 6} and strike {90, 95, 100%}; and five monetisation rules × four tenor structures, each also measured from 1 July 2020. The heatmap is fig8. Finally `json.dump(RESULTS, …)` writes every number used in the report and the deck.

10. The deck, slide by slide

17 slides in five sections, each with full speaker notes in the `.pptx`. At about a minute a slide the talk runs 15 to 18 minutes; slides 5, 9 and 13 are the easiest to shorten. Numbers come from `output/results.json` (keys in `code` font) or the named CSV.

1. **Title.** Three syllabus topics, then the election, then a hedge I designed and backtested.
2. **Four numbers.** Options overpriced by about 2 points; fear triples volatility; election puts jumped 4.2×; the hedge cut the worst fall from 38% to 22%. *From:* `vrp`, `sameday_by_regime`, `election.put22500_path`, `backtest_performance.csv`.

3. **Real prices, modelled premiums.** Index data is real and spot-checked; premiums are Black-Scholes at VIX + skew; regimes cover 27%, 48% and 25% of days. *From:* `sample.regime_share`.
4. **VIX usually overstates future volatility.** VIX beat realised volatility 78% of the time by 2.2 points (t = 2.8), and by 4 points after shocks; the red patches are crashes. *From:* fig2, `vrp_by_regime.csv`.
5. **How India VIX is built.** Model-free variance for two expiries, interpolated to 30 days; my replication gives 15.0 flat and 16.1 with skew. *From:* `vix_construction`.
6. **VIX as a fear gauge.** It rises 2.7% per 1% Nifty fall; regimes persist; spikes revert; it climbs before known events. *From:* fig1, `regimes`.
7. **Regimes separate risk, not return.** Volatility goes from 9% to 30%, returns do not differ (p = 0.66), and the worst 3 months started in Normal. *From:* `sameday_by_regime.csv`, `returns_by_regime.csv`.
8. **4 June 2024.** VIX doubled in six weeks, exit polls made puts cheapest the day before, Nifty fell 5.9% on result day, and VIX fell 29% the next day. *From:* fig4, `election`.
9. **Priced the size, not the direction.** Event variance implied 5.8% against an actual 5.9%; the election was the only shock with VIX already high. *From:* `event_comparison.csv`, `election.events`.
10. **The proposed hedge.** The five rules; 3 months because it decays slowly and survives a gap; tell the January 2020 trade. *From:* the `timed_hybrid` branch of `run()`.
11. **Backtest.** 13.9% against 11.1% a year, −22% against −38%, hedged 45% of the time for 2.9% a year in premium; the gap opens in March 2020. *From:* fig6, `backtest_performance.csv`.
12. **Always-on hedges cost more than they saved.** They cost 3% to 5% a year net; "hold" looks best only because of COVID; the hybrid guards against spikes that reverse. *From:* `backtest_performance.csv`.
13. **Calm-market purchases paid.** By regime at purchase, always-on hedges lost everywhere, while the proposal's low-VIX buys made 2.2% a year. *From:* `hedge_cost_payoff_by_regime.csv`, `excess_return_by_regime_annualised.csv`.
14. **Protection works from a calm start.** It works for sudden shocks from calm (2020, August 2024) and holds nothing in slow grinds (2022, late 2024, April 2025). *From:* `episode_returns.csv`.
15. **What the result depends on.** Robust: timed beats always in every cell. Fragile: tenor and the one crash; from July 2020 it returned 14.9% against 15.4%. Say this before they ask. *From:* fig8, `sensitivity_*.csv`, `ex_covid_performance`.
16. **Limitations.** Modelled premiums, one crash, synthetic expiries, no costs; the next step is 2008+ data and real option prices.

17. **Conclusion.** VRP about 2 points; VIX gauges fear, not direction; event premiums collapse in a day; the hedge is cheap crash insurance, not a return engine.

How `build_deck.js` makes it. The deck is built in code with **pptxgenjs** (a Node library that writes PowerPoint), 466 lines. `THEME` sets Cambria/Calibri and the colours (navy, orange for fear, blue for calm and the proposal). Three slide masters: `TITLE_DARK` (slide 1), `CONTENT` (2–16, with footer and slide number) and `CLOSING_DARK` (17). Helpers draw the repeated pieces: `card`, `numCircle`, `headDesc`, `stat` (big-number cards), `bullets`, `img` (keeps the PNG's aspect ratio), `barChart` (native, editable charts on slides 4, 7, 12, 13) and `tableRows`. Each slide is a few calls: `addSlide({ masterName })`, title into the placeholder, figures, text, `addNotes(...)`. **Numbers are typed in, not linked**: if you re-run the Python with new assumptions, update the slides by hand (or make the script read `results.json`).

To rebuild on your laptop: `npm install pptxgenjs`; change `FIG` (line 7) to your local `figures` folder; the `applyTheme` helper (line 5) is a project tool not on npm, so delete it and its call near the end (colours then fall back to Office defaults unless you swap in hex codes); run `node build_deck.js fdrm_vix_presentation.pptx`. The PDF came from `soffice --headless --convert-to pdf`, using Carlito and Caladea (metric-identical to Calibri and Cambria); exporting from PowerPoint avoids font differences.

11. Glossary

Term	Meaning
ATM, OTM	At / out of the money. A put is OTM when its strike is below spot.
Backtest	Replaying history day by day, applying a strategy's rules as if trading.
Bhavcopy	NSE's daily file of closing and settlement prices for every listed contract.
CAGR	Constant yearly return that turns the start value into the end value.
Calmar ratio	CAGR divided by the size of the maximum drawdown.
Collar	Buy a put, sell a call to pay for it: cheap protection, capped upside.
Convexity	Gains that accelerate as the move gets bigger; long options have it.

Term	Meaning
Data snooping	Choosing parameters on the same data used to test them, which flatters results.
Delta, gamma, vega, theta	Option sensitivity to spot, to changes in delta, to implied volatility, and to time passing.
Drawdown	Fall from the highest previous value; maximum drawdown is the worst one.
Event variance	Extra variance a known event adds, backed out of the rise in VIX before it.
Forward (F)	Fair future price of Nifty for an expiry: S × e^((r − q)T).
Implied volatility (IV)	Volatility that makes Black-Scholes match an option's market price.
Intrinsic value	Worth if exercised now; for a put, max(K − S, 0). Premium minus this is time value.
Leverage effect	Volatility rises more when prices fall than it falls when they rise.
Log return	ln(P_t ÷ P_t−1); adds up over time.
Look-ahead bias	Using information a backtest could not have known at the time.
Mean reversion	Drifting back towards a long-run average; VIX shows it strongly.
Model-free variance	Variance computed from a strip of option prices without a pricing model; the basis of VIX.
Monetise	Sell a hedge that has gained value.
Newey-West (HAC)	Standard errors valid when observations overlap or are autocorrelated.
Protective put	Owning an asset plus a put on it: a floor under losses.
Put spread	Buy a put, sell a lower-strike put: cheaper, but protection stops at the lower strike.
Realised volatility (RV)	Annualised standard deviation of actual returns over a window.

Term	Meaning
Regime	VIX band: Low below 13, Normal 13 to 20, High above 20.
Roll	Sell an option and buy a new one with a later expiry.
Sharpe, Sortino	Return above the risk-free rate divided by volatility (Sortino: downside volatility only).
Skew	IV differing by strike; for indices, OTM puts carry the highest IV.
Straddle	ATM call plus ATM put; its price measures the expected size of a move.
Tenor	Time from purchase to expiry; 3 months here.
Term structure	IV by expiry; normally upward-sloping, inverted in crises.
Vol crush	The sharp fall in IV once an event's outcome is known.
Vol point	One percentage point of annualised volatility (VIX 12 to 13).
VRP	Volatility risk premium: implied minus subsequently realised volatility.